

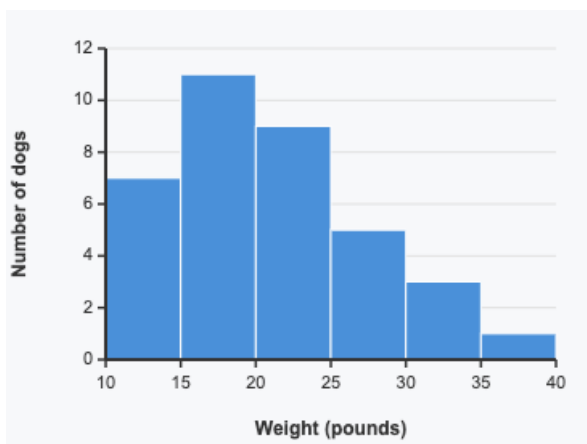
Lesson Objective

Create a histogram to represent a data set and interpret a histogram to answer questions about it.

Materials: Straightedge

1. Interpret a histogram to answer questions about a data set.

The histogram shows the weights of dogs at dog parks throughout a state. Each bar includes the left-end value but not the right-end value. For example, the first bar includes dogs that weigh 10 pounds and 15 pounds but not 20 pounds.



- How many dogs weigh at least 20 pounds and less than 40 pounds?
- How many dogs weigh exactly 32 pounds?
- How many dogs weigh at least 30 pounds?

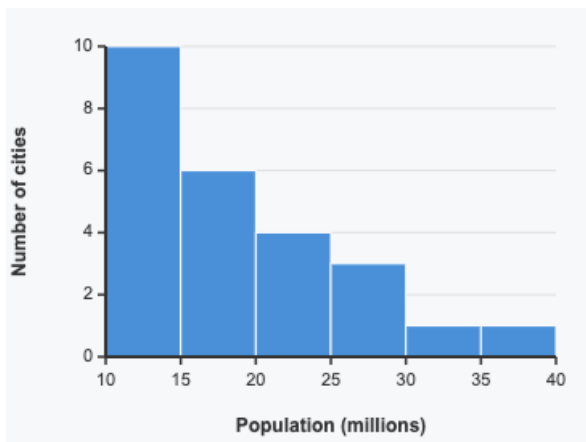
Show students the histogram and orient them to the axes. Point out that the x-axis groups dog weights into intervals of 5 pounds and the y-axis shows the number of dogs in each interval.

Remind students of the rule: the left-end value of each bar is included, but the right-end value is not. Use the first bar as an example: dogs weighing 10 pounds up to but not including 15 pounds.

For part a, guide students to identify the two bars that cover 20 to 40 pounds and add their heights: $11 + 9 + 5 = 25$ dogs. Note that 40 is not counted because the right-end value is excluded.

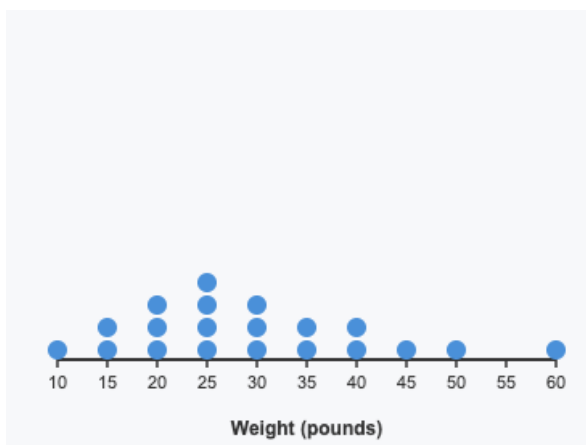
For part b, ask students what the bar from 30 to 35 pounds tells them. Point out that the histogram shows 5 dogs in that interval, but does not show the weight of any individual dog, so the count for exactly 32 pounds cannot be determined.

<i>Answer: a) 25 dogs; b) Cannot be determined from the histogram; c) 18 dogs</i>	For part c, have students add the heights of all bars from 30 pounds onward: $5 + 3 + 1 = 9$. Then guide them to include the bar from 25 to 30 by clarifying that "at least 30" starts at 30, so they should add: $9 + 9 = 18$ dogs.														
2. Create a histogram from a frequency table to represent a data set. The frequency table shows the populations of the 25 largest cities in the world, in millions. Use the table to create a histogram on the grid. <table data-bbox="224 869 781 1289"> <thead> <tr> <th>Population (millions)</th><th>Frequency</th></tr> </thead> <tbody> <tr> <td>10–15</td><td>10</td></tr> <tr> <td>15–20</td><td>6</td></tr> <tr> <td>20–25</td><td>4</td></tr> <tr> <td>25–30</td><td>3</td></tr> <tr> <td>30–35</td><td>1</td></tr> <tr> <td>35–40</td><td>1</td></tr> </tbody> </table> <i>Answer:</i>	Population (millions)	Frequency	10–15	10	15–20	6	20–25	4	25–30	3	30–35	1	35–40	1	Tell students they will use the frequency table to draw a histogram. Each bar will represent one interval, and the height of the bar will match the frequency in the table. Ask students where the first bar should start on the x-axis and how tall it should be. Guide them to draw the first bar from 10 to 15 with a height of 10. Have students draw the remaining bars one at a time, using a straightedge. Check that each bar covers the correct interval on the x-axis and reaches the correct frequency on the y-axis. As students are ready, have them finish the histogram independently. Point out that there are no gaps between bars because the intervals are contiguous.
Population (millions)	Frequency														
10–15	10														
15–20	6														
20–25	4														
25–30	3														
30–35	1														
35–40	1														



3. Compare a dot plot and a histogram to describe a typical value and the distribution of data.

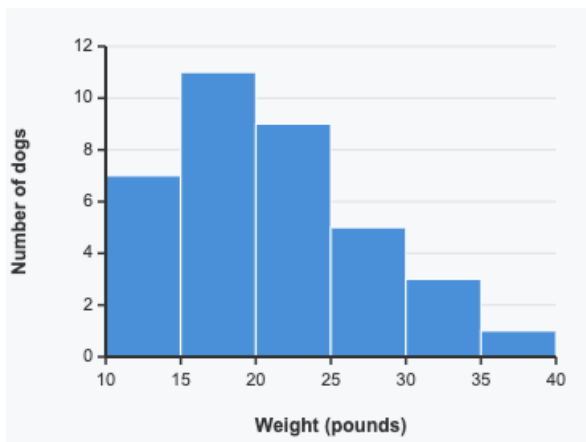
The dot plot and histogram both show the weights of dogs at dog parks. Use them to answer the questions.



Students work independently on parts a, b, and c.

After students finish, ask them to justify the typical value they chose by pointing to the interval on the histogram with the most dogs. Confirm that a typical value sits near the center of where the data is concentrated.

Ask students to share one advantage of the histogram. Reinforce that the histogram groups data into intervals, which makes the overall shape of a large data set easier to read, even though it hides individual values.



- What is a typical weight of a dog? Explain how you know.
- Describe the distribution of dog weights.
- Name one advantage of using a histogram.

Answer: a) A typical weight is between 15 and 20 pounds, because that interval has the most dogs and the bulk of the data is concentrated in the lower weights; b) Most dogs weigh between 10 and 25 pounds. Fewer dogs are in the heavier intervals, and the number of dogs decreases as the weight increases; c) A histogram makes it easier to see the overall shape of a large data set at a glance.

Monitor For

Instructional Moves

• Do students correctly apply the left-included, right-excluded rule when reading bar intervals in Problem 1? • Do students recognize that a histogram cannot tell them how many data values fall at one exact number (Problem 1, part b)? • Do students match each bar's height in Problem 2 to the correct frequency on the y-axis, especially for frequencies that fall between gridlines? • Do students identify a typical value from the interval with the greatest frequency rather than from a single data point in Problem 3?	• Before students answer Problem 1, trace the first bar with a finger and say, "This bar covers 10 pounds up to but not including 15 pounds." Repeat with one more bar to anchor the convention. • If a student counts only one bar for "at least 20 and less than 40," ask, "Which intervals fall between 20 and 40?" to prompt them to add both bar heights. • When drawing bars in Problem 2, have students place a small mark on the y-axis at the correct frequency first, then draw the bar up to that mark with a straightedge. • If a student names a single weight as the typical value in Problem 3, ask, "Where is most of the data concentrated?" to redirect them to the interval with the greatest frequency.
Reflection Questions • In Problem 1, how did you know which bars to add to find the number of dogs weighing at least 20 and less than 40 pounds? <i>Answer: I added the heights of the bar from 20 to 25 and the bar from 25 to 30, and so on, stopping before 40 because the right-end value is not included.</i> • Why could you not tell from the histogram how many dogs weigh exactly 32 pounds? <i>Answer: The bar shows the total number of dogs in the interval from 30 to 35 pounds, but it doesn't show the weight of any individual dog.</i>	

- When you drew the histogram in Problem 2, how did the frequency in the table tell you what to draw? *Answer: The frequency told me how tall to make each bar on the y-axis.*

- What is one advantage of a histogram compared to a dot plot? *Answer: A histogram makes it easier to see the overall shape and spread of a large data set, even though it doesn't show individual values.*